THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS IX — MATHEMATICS · FULL SYLLABUS EXAMINATION

Session 2026-27 | CBSE / RBSE Pattern

Class: 9	Subject: Maths	Set 01	Time: 3 Hours	Max. Marks: 80

General Instructions:

1. This question paper contains 38 questions and is divided into 5 sections — A, B, C, D and E.
2. In Section A, Q. 1–18 are multiple-choice questions and Q. 19–20 are Assertion–Reason based questions of 1 mark each.
3. In Section B, Q. 21–25 are very short answer (VSA) type questions carrying 2 marks each.
4. In Section C, Q. 26–31 are short answer (SA) type questions carrying 3 marks each.
5. In Section D, Q. 32–35 are long answer (LA) type questions carrying 5 marks each.
6. In Section E, Q. 36–38 are case-study based questions carrying 4 marks each with sub-parts of 1, 1 and 2 marks respectively.
7. All questions are compulsory. However, an internal choice is provided in 1 question of Section B, 2 questions of Section C and 2 questions of Section D. An internal choice is provided in all the 2-mark sub-parts of Section E.
8. Draw neat and clean figures wherever required. Take π = 22/7 wherever required, if not stated. Use of calculators is not allowed.
9. Write the serial number of each question before attempting it.

SECTION A — Objective Type — 20 questions · 20 marks

Q1–Q18 are multiple-choice questions (1 mark each). Q19–Q20 are Assertion (A) and Reason (R) questions: select the correct option from (a), (b), (c), (d) given below each.

1. The point (−5, 0) lies on the: **1**

(a) y-axis (b) x-axis
(c) third quadrant (d) second quadrant

2. Which of the following can be the coordinates of a point in the fourth quadrant? **1**

(a) (3, −2) (b) (−3, −2) (c) (−3, 2) (d) (3, 2)

3. The midpoint of the line segment joining A(2, −6) and B(−4, 10) is: **1**

(a) (1, 2) (b) (−1, −2) (c) (−1, 2) (d) (−3, 8)

4. A rectangle has length (2x + 1) cm and breadth (x − 3) cm. Its perimeter (in cm) is: **1**

(a) 6x + 4 (b) 3x − 2 (c) 3x + 4 (d) 6x − 4

5. The value of the polynomial $p(x) = 3x^2 - 4x + 1$ at x = −1 is: **1**

(a) 8 (b) 0 (c) −6 (d) 2

6. A cab company charges ₹40 as fixed charge and ₹14 per km. The fare (in ₹) for d km is: **1**

(a) 40d + 14 (b) 14d + 40 (c) 54d (d) 14d − 40

7. The absolute value of the rational number $\frac{-7}{9}$ is: **1**

(a) $\frac{-7}{9}$ (b) $\frac{-9}{7}$ (c) $\frac{7}{9}$ (d) $\frac{9}{7}$

8. Which of the following is an irrational number? **1**

(a) 5.0505…(0.05 repeating) (b) √0.04
(c) √18 (d) √(81/25)

9. Which of the following has a terminating decimal expansion? **1**

(a) $\frac{5}{12}$ (b) $\frac{4}{15}$ (c) $\frac{11}{18}$ (d) $\frac{9}{80}$

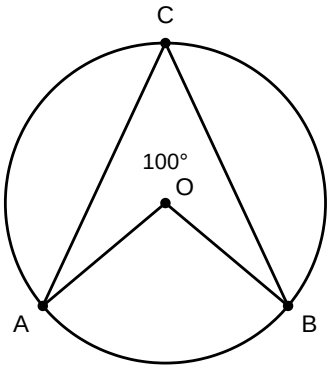
10. On expanding $(3a - 2b)^2$, we get: **1**

(a) $9a^2 - 12ab + 4b^2$ (b) $9a^2 - 6ab + 4b^2$
(c) $9a^2 - 12ab - 4b^2$ (d) $9a^2 + 12ab + 4b^2$

11. Using a suitable identity, the value of 98×102 is: **1**

(a) 9996 (b) 10004 (c) 9998 (d) 9904

12. In the figure, O is the centre of the circle and $\angle AOB = 100°$. Then $\angle ACB$ is equal to: **1**



(a) 200° (b) 100° (c) 50° (d) 25°

13. A chord of length 16 cm is drawn in a circle of radius 10 cm. Its distance from the centre is: **1**

(a) $\sqrt{164}$ cm (b) 4 cm (c) 6 cm (d) 8 cm

14. The sides of a triangle are in the ratio 4 : 5 : 6 and its perimeter is 90 cm. The shortest side is: **1**

(a) 15 cm (b) 30 cm (c) 20 cm (d) 24 cm

15. The area of a triangle with sides 5 cm, 12 cm and 13 cm, found using Heron's formula, is: **1**

(a) 78 cm² (b) 32.5 cm² (c) 30 cm² (d) 60 cm²

16. A letter is chosen at random from the word 'EDUCATION'. The probability that it is a vowel is: **1**

(a) $\frac{4}{9}$ (b) $\frac{1}{2}$ (c) $\frac{5}{8}$ (d) $\frac{5}{9}$

17. A die is thrown once. The probability of getting a number less than 5 is: **1**

(a) $\frac{1}{3}$ (b) $\frac{5}{6}$ (c) $\frac{1}{2}$ (d) $\frac{2}{3}$

18. If the n-th term of a sequence is $t_n = n^2 - 3n$, then t_7 is: **1**

(a) 70 (b) −28 (c) 14 (d) 28

19. **Assertion (A):** $\sqrt{2} + 1$ is an irrational number. **1**
Reason (R): The sum of a rational number and an irrational number is always irrational.
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

20. **Assertion (A):** For the sequence defined by $t_1 = 2$ and $t_n = t_{n-1} + 4$ $(n \geq 2)$, the third term is 12. **1**
Reason (R): A recursive rule gives every term from the term before it.
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION B — Very Short Answer (2 marks each) — 5 questions · 10 marks

Internal choice is provided in one question.

21. A taxi charges ₹50 for the first 3 km and ₹18 for every km after that. What is the fare for a journey of 12 km? **2**

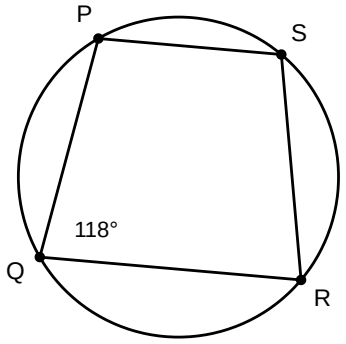
22. Without actual division, decide whether the decimal expansion of $\frac{17}{320}$ is terminating. If it is, state the number of decimal places. **2**

OR

Find any two rational numbers between $\frac{1}{4}$ and $\frac{1}{3}$.

23. Factorise: $9a^2 + 30ab + 25b^2$. **2**

24. In the figure, PQRS is a cyclic quadrilateral with $\angle PQR = 118°$. Find $\angle PSR$ and state the reason. **2**



25. Find the second, third and fourth terms of the sequence given by $t_1 = 3$ and $t_n = 2t_{n-1} - 1$ for $n \geq 2$. **2**

SECTION C — Short Answer (3 marks each) — 6 questions · 18 marks

Internal choice is provided in two questions.

26. Show that $0.153846153846\ldots = \frac{2}{13}$. **3**

27. A school canteen recorded the drink chosen by 100 students in a day: **3**

Drink	Tea	Coffee	Juice	Milk
Students	30	20	35	15

A student is selected at random. Find the probability that the student chose (i) juice, (ii) tea or milk, (iii) not coffee.

OR

In a survey of 80 households the number of members was recorded:

Members	2	3	4	5
Households	10	25	30	15

A household is chosen at random. Find the probability that it has (i) exactly 4 members, (ii) at most 3 members, (iii) more than 4 members.

28. Write the coordinates of the point (i) which lies on the x-axis, 7 units to the left of the origin; (ii) which lies on the y-axis, 4 units below the origin; (iii) which is the mirror image of (−3, 5) in the x-axis. **3**

29. Simplify: $(x + 2)^3 + (x - 2)^3$. **3**

30. A chord of length 30 cm is drawn in a circle of diameter 34 cm. Find (i) the distance of the chord from the centre; (ii) the length of another chord of this circle which is at a distance of 15 cm from the centre. **3**

31. Using Heron’s formula, find the area of a triangle whose sides are 17 cm, 25 cm and 26 cm. **3**

OR

The sides of a triangle are in the ratio 3 : 4 : 5 and its perimeter is 72 cm. Find its area.

SECTION D — Long Answer (5 marks each) — 4 questions · 20 marks

Internal choice is provided in two questions.

32. An architect marks the four corners of a rectangular courtyard on a grid as A(−2, −1), B(4, −1), C(4, 3) and D(−2, 3). **5**

(a) Plot the points on a coordinate plane and join them in order. (2 marks)

(b) Find the lengths AB, BC, CD and DA. (1½ marks)

(c) Using the diagonals, justify that ABCD is a rectangle but not a square. (1½ marks)

33. How many 3-digit numbers leave remainder 2 when divided by 6? Find the sum of all such numbers. (2½ + 2½) **5**

34. A battery company tested 500 batteries and recorded their life: **5**

Life (hours)	0 – 50	50 – 100	100 – 150	150 – 200
Number of batteries	20	70	190	220

Find the probability that a randomly chosen battery lasts (i) less than 50 hours (1), (ii) between 50 and 150 hours (2), (iii) more than 150 hours (2).

OR

A box contains 6 blue pens, 8 black pens and 4 red pens. A pen is drawn at random. Find the probability that it is (i) blue (1), (ii) black or red (2), (iii) neither blue nor black (2).

35. The length of a rectangular play area is 5 m more than its breadth, and its area is 84 m². Form a quadratic expression in the breadth x, factorise it and find the perimeter of the play area. **5**

OR

A carpentry student uses 3 large square tiles of side x units, 10 rectangular strips of length x and width 1 unit, and 7 small square tiles of side 1 unit to make one large rectangle without overlap. Find the length and breadth of the rectangle in terms of x.

SECTION E — Case-Study Based Questions (4 marks each) — 3 questions · 12 marks

Read each case carefully and answer the questions that follow. Internal choice is provided in every 2-mark sub-part.

36. Priya's mobile data pack gives her 60 GB at the start. She uses 4 GB of data every day. The data (in GB) left after x days is given by D = 60 − 4x.

(i) How much data is left after 5 days? 1

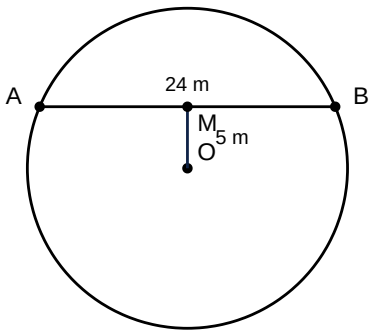
(ii) State the degree of D = 60 − 4x and the coefficient of x. 1

(iii) After how many days will only 12 GB of data be left? 2

OR

(iii) After how many days will her data pack be exhausted?

37. A circular pond with centre O is built in a park. A fence AB, 24 m long, is fixed as a chord of the pond. The perpendicular distance of the fence from the centre, OM, is 5 m.



(i) Find the radius of the pond. 1

(ii) Another fence CD, also 24 m long, is fixed in the same pond. How far is it from the centre? 1

(iii) A walking path PQ is a chord at a distance of 12 m from O. Find its length. 2

OR

(iii) Four lamps are placed at A, B, D and E on the boundary forming a cyclic quadrilateral ABDE. If $\angle ADE = 75°$, find $\angle ABE$ and give the reason.

38. A residents' association has a triangular garden whose sides are 10 m, 17 m and 21 m.

(i) Find the semi-perimeter of the garden. 1

(ii) Find the area of the garden using Heron's formula. 1

(iii) Find the length of the altitude drawn to the longest side. 2

OR

(iii) Grass is to be laid over the whole garden at ₹25 per m^2. Find the total cost.